Structuring Judgment Under Constraint: Cognitive Friction in Procedural Texts

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Abstract Procedural documents such as Standard Operating Procedures (SOPs) actively structure action and accountability in regulated work, yet prior research has not theorized how document structure — rather than individual cognition — produces localized judgment demands during execution. This study introduces **cognitive friction** as a document-centered framework for analyzing how procedural texts embed structurally produced interpretive demands within routine instructions. Drawing on a corpus of 500 procedural sentences derived from publicly available regulatory and safety documentation, the analysis operationalizes cognitive friction as the co-occurrence of four structural features: conditional branching, cross-references, exception structures, and discretionary qualifiers, using rule-based computational annotation to identify feature distributions and co-occurrence patterns across the corpus. Results show that individual friction features are common across procedural genres, but structurally compounded judgment demands concentrate within a smaller subset of sentences where conditional structures co-occur with referential or discretionary cues — a pattern that holds consistently across naturally occurring procedural documentation, confirming that structurally embedded judgment demands are a general property of procedural text architecture rather than an artifact of any single institutional genre. By demonstrating that responsibility-relevant judgment demands can be operationalized and observed at the level of procedural sentence structure, these findings position cognitive friction as the procedural stage of responsibility production within a broader multi-stage model spanning anticipatory alignment, investigative stabilization, and representational mediation, with direct implications for the design of procedural documents and interactive systems that mediate how actors navigate instructions under accountability constraints.

Keywords: cognitive friction, procedural texts, standard operating procedures, accountability, judgment demands, regulated work, document structure

Introduction

Organizational procedures in regulated environments are enacted through written documents: Standard Operating Procedures (SOPs), protocols, and compliance manuals (Berg, 1997). These procedural texts do not merely record organizational knowledge; they structure how actors interpret situations, sequence actions, and allocate responsibility under conditions of

uncertainty (Suchman, 1987; Hutchins, 1995; Pentland & Feldman, 2005). Across domains such as healthcare, laboratory science, and industrial production, organizational action is routinely mediated through such documents, making procedural texts central infrastructures for organizing judgment and accountability in practice.

Prior scholarship has largely treated procedural documents as carriers of standardized knowledge or coordination devices. This framing captures how procedural texts transmit instruction and align action across organizational actors — but does not address how procedural language itself structures the interpretive burden that actors must resolve during execution. Specifically, existing accounts have not systematically addressed how document structure determines where localized judgment becomes necessary and how that judgment remains individually attributable even when its necessity originates in textual design.

Existing accounts of cognitive difficulty in organizational settings predominantly locate breakdown, strain, and error in human limitations (Reason, 1990; Norman, 1988). Whether framed in terms of information-processing load, task complexity, or executional failure, analysis centers on the capacities and actions of individuals. Within this orientation, formal documents are positioned as stable and neutral artifacts that transmit instruction, while responsibility for successful enactment rests primarily with the actor. As a result, how document structure shapes interpretive burden — and thereby stabilizes responsibility-relevant judgment conditions — remains insufficiently theorized in organizational research (Heimer, 2012; Vaughan, 1996).

This study introduces **cognitive friction** in procedural texts, defined as structural features of documents that increase the interpretive effort required to translate instructions into situated action and judgment, thereby shaping how responsibility is allocated during procedural work. Rather than viewing procedural difficulty as an individual cognitive limitation, cognitive friction conceptualizes difficulty as emerging from document organization — such as conditional branching, deferred reference, and exception layering — that structures how actors navigate instructions and justify decisions. Cognitive friction thus refers not to task complexity or informational load per se, but to the interpretive demands produced by procedural text structure.

Procedural texts not only guide action but also structure how responsibility becomes attributable in regulated organizational work. When procedural instructions embed contingent pathways, deferred criteria, or discretionary qualifiers, actors must resolve locally situated interpretations while remaining individually accountable for resulting action. Responsibility in such settings is therefore not solely stabilized through retrospective evaluation or institutional attribution processes — it is partially pre-structured within the architecture of procedural language itself. By specifying how document structure organizes localized judgment demands, cognitive friction identifies a document-level mechanism through which procedural texts participate in the sociotechnical production of organizational accountability.

The analysis draws on a corpus of 500 procedural sentences derived from publicly available

regulatory and safety documentation, including procedural guidance documents and laboratory safety manuals. By examining how friction features distribute and co-occur across procedural sentences, this study develops a document-centered account of how procedural texts structure responsibility through localized judgment demands.

This study makes three contributions. First, it conceptualizes cognitive friction as a document-level property that structures interpretive burden in procedural work. Second, it provides a corpus-based demonstration that structurally embedded judgment demands are observable and measurable within procedural sentence architecture — an operationalization that prior STS and organizational accounts have theorized but not systematically specified. Third, it demonstrates how document structure participates in the stabilization of responsibility in organizational work, situating cognitive friction as the procedural stage within a broader multi-stage model of responsibility production spanning anticipatory alignment (AAC), procedural embedding (CF), investigative stabilization, and representational mediation. These contributions have direct implications for the design of procedural documents and interactive systems through which actors navigate instructions under accountability constraints.

Conceptual Framing

Accounts of cognitive difficulty in organizational settings predominantly locate breakdown, strain, and error in human limitations (Reason, 1990; Norman, 1988). Whether framed in terms of information-processing load, task complexity, or executional failure, analysis centers on the capacities and actions of individuals. Within this orientation, formal documents are positioned as stable and neutral artifacts that transmit instruction, while responsibility for successful enactment rests primarily with the actor.

Such an analytical stance becomes unstable in highly regulated environments, where documentation does more than describe tasks. Procedural texts define the conditions under which interpretation becomes necessary, specify (or defer) criteria for decision-making, and establish the grounds upon which actions will later be evaluated. Judgment in these settings is not merely a moment of cognition but an event that is recorded, reviewed, and retrospectively scrutinized. Under these conditions, responsibility cannot be understood solely as an attribute of individual performance; it must also be examined as an effect of how judgment is structured, distributed, and stabilized within documentation practices (Suchman, 1987; Weick, 1995).

This study introduces the concept of **cognitive friction** to render structurally produced judgment demands analytically visible. Cognitive friction captures moments in which procedural language requires local interpretation, reconciliation, or discretionary adjustment without fully specifying the criteria for doing so. These moments are consequential because the resulting judgments remain individually attributable, even when their necessity arises from textual design

rather than actor discretion.

Obligation cues — modal or prescriptive constructions such as *shall*, *must*, and *required to* — warrant particular attention here. Although obligation cues signal mandated action and are ubiquitous in procedural language, they function as the prescriptive baseline of procedural texts rather than as sources of structurally contingent interpretive demand. Cognitive friction arises not from obligation itself, but from conditions layered on top of it: contingent pathways, deferred criteria, and discretionary qualifiers that leave interpretive resolution partially unspecified. Obligation cues are therefore retained as an annotated feature for descriptive purposes but excluded from the composite cognitive friction score, which is reserved for features that generate structurally contingent judgment demands.

To observe these dynamics at a granular level, this study adopts the **sentence** as the minimal unit of analysis. The sentence is theoretically motivated as the unit of procedural inscription at which accountability-relevant judgment becomes locally attributable: procedural sentences are the smallest syntactic units that simultaneously specify an action, its conditions of application, and the interpretive resolution required to execute it. Higher-order units such as paragraphs or sections distribute interpretive work across multiple steps and actors, while sub-sentential units do not encode the full conditional or discretionary structure through which judgment demands are produced. Individual procedural sentences frequently encode conditional branching, obligation cues, cross-references, and exception handling within a single syntactic structure — making the sentence the appropriate site at which to observe the co-occurrence of features that produce localized interpretive demand. By examining how such features co-occur, the framework renders visible how documentation practices organize judgment and, in doing so, participate in the production of responsibility.

The analysis is situated in the broader genre of regulated procedural documentation, particularly SOP-like instructional texts used in compliance and safety-critical environments. Such procedural documents combine prescriptive authority, auditability, and executional traceability within a single documentary form. Their sentences do not merely describe recommended action but define accountable action under conditions of regulatory scrutiny. This artifact specificity is critical: cognitive friction in procedural texts emerges not simply from linguistic complexity but from the requirement that procedural language simultaneously prescribe action, anticipate exceptions, and sustain audit-relevant justification. The framework therefore treats procedural sentences as sociotechnical artifacts in which interpretive demands and responsibility conditions are co-encoded.

This genre orientation allows the analysis to extend beyond domain-specific SOPs to procedurally structured texts across regulatory and safety contexts. Research in science and technology studies has long emphasized that inscriptions and formal representations do not merely reflect practice but participate in structuring it — stabilizing categories, distributing interpretive work,

and rendering action accountable within institutional settings (Bowker & Star, 1999; Latour, 1987). The present analysis builds on this tradition by specifying how such structuring occurs at the level of procedural sentence architecture, where conditional, referential, and obligation structures organize localized judgment demands within rule-governed work.

Within organizational theory, documentation has similarly been examined as a mechanism of standardization and control, particularly in contexts of formalized compliance and auditable practice (Power, 1997; Orlikowski, 2007). Prior work has treated procedural texts as carriers of prescribed action rather than as sites where interpretive resolution becomes structurally necessary. By introducing cognitive friction as an observable and measurable property of procedural language, this study shifts attention toward how documentation distributes decision criteria and allocates responsibility-relevant judgment to actors during execution.

Cognitive friction extends sociotechnical perspectives on accountability by linking textual design to the stabilization of attributable action. Rather than locating responsibility solely in retrospective evaluation or actor discretion, the framework highlights how procedural inscriptions pre-structure the conditions under which judgment becomes necessary and individually attributable — extending existing theories of documentation and organizational coordination by demonstrating how responsibility is partially organized within the architecture of procedural text itself. This has direct implications for the design of procedural documents and interactive systems that support actors in navigating instructions under accountability constraints.

Methods: Corpus Construction and Computational Annotation

This study analyzes how structurally embedded judgment demands are distributed across procedural documentation using a mixed corpus of synthetic and naturally occurring sentences, balancing structural control with ecological validity in procedural text analysis.

The corpus consists of 500 procedural sentences sampled from three sources: synthetic SOP-like sentences designed to isolate common structural configurations of procedural language, publicly available procedural guidance documents, and laboratory safety and compliance manuals used in regulated operational environments. Of these, 100 sentences were synthetically constructed to isolate recurrent procedural structures, and 400 were derived from publicly accessible procedural and regulatory documentation. Together, these sources represent procedural documentation across regulatory and safety contexts while preserving analytic clarity of structural features.

The inclusion of synthetic sentences serves a methodological purpose rather than a substitution for empirical material. Naturally occurring procedural corpora present a structural

observation problem: high-friction configurations — sentences combining conditional branching with cross-references or discretionary qualifiers — are comparatively rare in uncontrolled documentary corpora, making it difficult to isolate and characterize their structural properties from occurrence frequency data alone. The synthetic subset addresses this directly by constructing sentences that exemplify target structural configurations, enabling systematic examination of feature co-occurrence patterns that would otherwise be underrepresented in a purely naturalistic sample. The synthetic component is therefore explicitly designed as an analytic control, not as a representation of real-world prevalence. Synthetic sentences support observation of structurally embedded judgment demands independent of domain-specific variation — functioning as an analytic control that isolates structural configurations otherwise difficult to observe in uncontrolled organizational corpora. To verify that findings are not driven by the synthetic subset, key analyses were replicated on the 400 naturally occurring sentences alone; results are reported alongside full-corpus findings in the Results section.

Procedural text is segmented at the sentence level, consistent with the theoretical framing of cognitive friction as a property of procedural inscription embedded within localized syntactic structure. Each sentence is evaluated for the presence of five structural features associated with judgment demands: obligation cues, conditional branching, cross-references, exception structures, and discretionary qualifiers.

Obligation cues were excluded from the composite score as they represent the prescriptive baseline of procedural language rather than contingent sources of interpretive demand. Cognitive friction arises when execution pathways, criteria, or judgment conditions remain partially specified on top of obligatory instruction — not from the presence of obligation per se. Feature detection is implemented using a rule-based computational annotation procedure to ensure reproducibility and consistency across the corpus.

Based on this annotation scheme, a composite cognitive friction score is computed for each sentence as the count of co-occurring friction features (conditional branching, cross-references, exception structures, and discretionary qualifiers). This operationalization captures localized interpretive demand as the within-sentence co-occurrence of friction features.

Annotation and Coding Procedure

Annotation of cognitive friction features was conducted at the sentence level using a rule-guided coding framework derived from the conceptual operationalization. Each sentence was evaluated for the presence of five structural features: obligation cues, conditional branching, cross-references, exception structures, and discretionary qualifiers. Coding criteria were specified to ensure interpretive consistency while preserving sensitivity to structural variation within procedural language.

Obligation cues were identified through modal or prescriptive constructions (e.g., *shall*, *must*, *required to*) indicating mandated action. Conditional branching was coded when sentences contained explicit conditional structures (e.g., *if*, *when*, *unless*) that introduced contingent procedural pathways. Cross-references were marked where sentences directed actors to external procedural elements, such as other sections, documents, or criteria sources. Exception structures were coded when language explicitly signaled deviation from default procedural flow (e.g., *however*, *except*, *unless otherwise specified*). Discretionary qualifiers were coded when sentences introduced actor-facing flexibility or justificatory latitude (e.g., *may*, *as appropriate*, *if needed*, *based on judgment*), thereby allocating interpretive resolution to the actor during execution.

Annotation was implemented as a rule-based computational procedure applied uniformly across the full corpus. All annotation rules are explicitly defined and consistently applied, ensuring full reproducibility of feature detection. The complete annotation scripts and an illustrative sentence subset are publicly available at `https://github.com/sorakim-lab/sop-cognitive-friction-pilot`, supporting transparency and replication.

Coding was conducted to capture structural presence rather than interpretive outcome. The objective was not to evaluate correctness of action or decision quality but to identify textual configurations that allocate interpretive resolution to actors during execution. This structural coding approach enables corpus-level aggregation while maintaining alignment with the theoretical definition of cognitive friction as a property of procedural inscription rather than individual cognition.

Synthetic Component and Structural Control

In addition to naturally occurring procedural sentences, the corpus includes a synthetic subset of 100 sentences deliberately constructed to isolate structural features associated with judgment demands. Each sentence was annotated using the same rule-based feature scheme applied to the empirical corpus, enabling direct structural comparison across synthetic and naturally occurring procedural texts.

The synthetic subset exhibits higher mean cognitive friction (M = 0.95) than naturally occurring sources (M = 0.47–0.60), consistent with its design purpose. Core analyses were replicated on the 400 naturally occurring sentences alone to confirm that this difference does not materially alter substantive findings. Feature frequencies remain stable for cross-references (11.8%) and discretionary qualifiers (13.2%), while conditional branching (26.2%) and exception structures (1.5%) are attenuated relative to the full corpus. The friction score distribution holds, with 38.8% of naturally occurring sentences containing a single friction feature — identical to the full-corpus rate — and co-occurrence of two or more features occurring in 7.0% of naturally occurring sentences compared to 11.0% in the full corpus. These results confirm that structurally

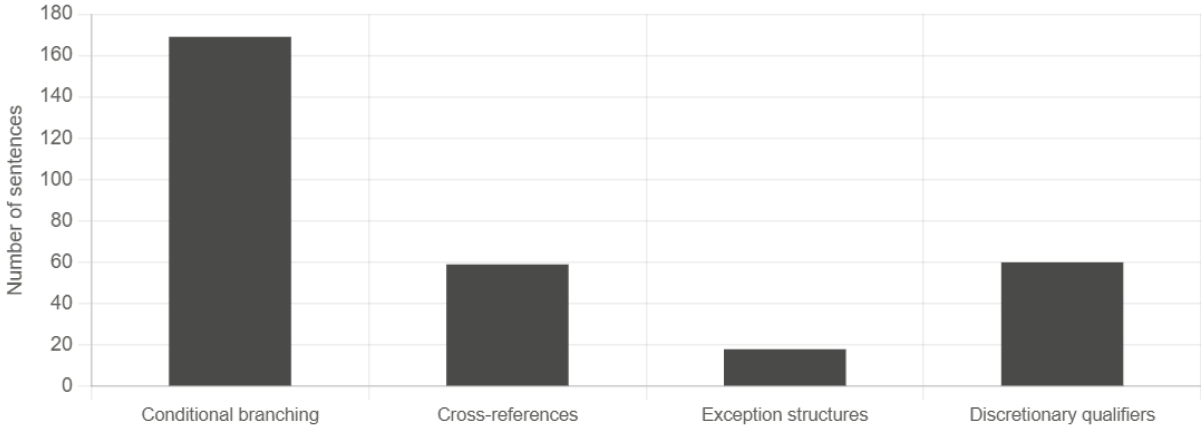


Figure 1: Frequency of cognitive friction features across procedural sentences in the empirical corpus (N = 500). Bars indicate the number of sentences containing each friction feature (conditional branching, cross-references, exception structures, discretionary qualifiers).

embedded judgment demands are not an artifact of synthetic construction but a general property of procedural text architecture across institutional genres.

Results

Feature Distribution

The empirical corpus consists of 500 procedural sentences drawn from publicly available regulatory and procedural documentation, including WHO guidance, biosafety protocols, chemical safety procedures, and synthetic SOP-like constructions. All sentences were annotated using the rule-based computational procedure described above.

Across the full corpus, conditional branching was the most prevalent cognitive friction feature, occurring in 33.8% of sentences, indicating that contingent execution pathways are routinely embedded at the sentence level of procedural documentation. Discretionary qualifiers appeared in 12.0% of sentences, while cross-references occurred in 11.8%. Exception structures appeared in 3.6% of sentences. Replication on the 400 naturally occurring sentences alone shows that these distributions remain stable for cross-references (11.8%) and discretionary qualifiers (13.2%), while conditional branching (26.2%) and exception structures (1.5%) are attenuated — consistent with the synthetic subset's design purpose of isolating high-friction structural configurations. These distributions indicate that procedural documentation frequently embeds contingent execution pathways and discretionary judgment conditions within individual sentences across both the full corpus and the naturally occurring subset.

Because cognitive friction is defined as the co-occurrence of structurally embedded judgment demands, feature combinations were examined at the sentence level. Across the full corpus, the majority of sentences contained no friction features (50.2%), while 38.8% contained a single

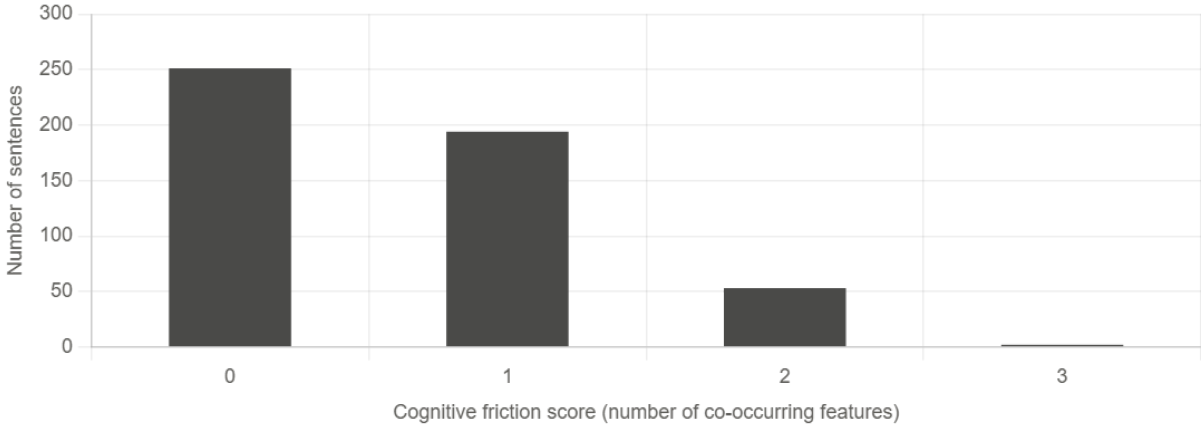


Figure 2: Distribution of composite cognitive friction scores across procedural sentences in the empirical corpus (N = 500). Bars indicate the proportion of sentences containing zero, one, or two or more co-occurring friction features. This figure presents an aggregated count-based distribution; pairwise co-occurrence structure is shown separately in Figure 3.

feature. Co-occurrence of two features occurred in 10.6% of sentences, and only 0.4% exhibited three simultaneous friction features. In the naturally occurring subset, the pattern holds: 54.2% of sentences contained no friction features, 38.8% contained a single feature, and 7.0% exhibited co-occurrence of two or more features. This consistency across both the full corpus and the naturally occurring subset confirms that while individual friction cues are common, structurally compounded judgment demands are comparatively concentrated within a smaller subset of procedural sentences — and that this concentration is not an artifact of the synthetic component.

Composite Cognitive Friction Across Procedural Sources

To examine whether cognitive friction varies across procedural genres, mean composite friction scores were compared across source categories. Sentences derived from synthetic SOP-like constructions exhibited the highest average friction (M = 0.95), followed by chemical safety procedures (M = 0.60), WHO guidance texts (M = 0.50), and biosafety protocols (M = 0.47).

This gradient suggests that documents designed to simulate tightly specified procedural execution contexts concentrate structurally embedded judgment demands more densely than regulatory or guidance-oriented texts. In contrast, institutional guidance documents, while prescriptive, appear to distribute interpretive demands more diffusely across procedural context rather than within individual sentences.

Importantly, the presence of non-zero friction across all naturally occurring source categories — BSL3 (M = 0.47), WHO (M = 0.50), chemical (M = 0.60) — confirms that cognitive friction is not an artifact of synthetic construction but emerges consistently across diverse forms of regulated procedural documentation. This supports the conceptual claim that structurally embedded judgment demands are a general property of procedural text architecture rather than

an artifact of any single institutional genre.

Structural Concentration of High-Friction Sentences

Inspection of the highest-scoring sentences reveals that elevated cognitive friction arises when conditional branching co-occurs with cross-referential or discretionary structures within a single procedural statement. Such sentences simultaneously specify contingent execution conditions while deferring decision criteria to external references or localized judgment. This configuration produces concentrated interpretive demand because actors must resolve execution pathways under conditions of partial specification while remaining individually accountable for the resulting action.

Although such high-friction sentences are numerically rare, their structural configuration illustrates the mechanism through which procedural language can organize locally intensified judgment demands. These observations align with the conceptual definition of cognitive friction as structurally produced moments of interpretation embedded within procedural inscription.

Illustrative High-Friction Procedural Configurations

To illustrate how cognitive friction emerges in procedural text architecture, representative high-friction sentences were examined qualitatively. These examples demonstrate how co-occurring structural features concentrate interpretive demand within a single procedural statement.

Example 1 — Conditional + Cross-reference + Discretion

If reporting is delayed, the analyst shall notify the supervisor in accordance with SOP-QA-017 as deemed necessary.

This sentence simultaneously introduces a contingent execution condition (*If reporting is delayed*), defers action specification to an external procedural reference (*in accordance with SOP-QA-017*), and embeds discretionary judgment (*as deemed necessary*). The actor must therefore determine both the applicability of the referenced SOP and the threshold for discretionary escalation, producing locally concentrated interpretive demand.

Example 2 — Conditional + Exception

When contamination is confirmed, the operator must continue processing except when safety criteria are compromised.

Here, a conditional trigger (*When contamination is confirmed*) is combined with an embedded exception structure (*except when safety criteria are compromised*). Execution requires the actor to reconcile competing procedural directives — continuation versus interruption — based on locally evaluated criteria. Responsibility for this reconciliation remains individually attributable despite originating from textual structure.

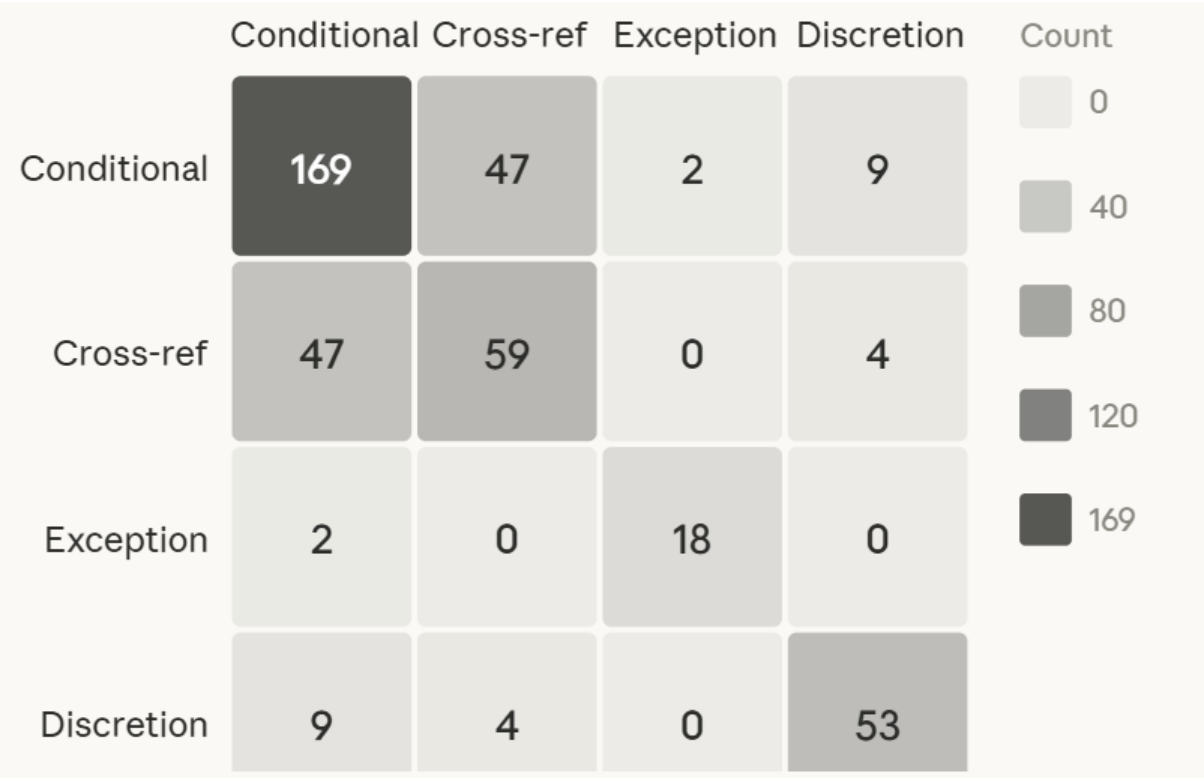


Figure 3: Pairwise co-occurrence matrix of cognitive friction features across procedural sentences in the empirical corpus (N = 500). Cells indicate the number of sentences in which each pair of friction features co-occurs within the same procedural statement.

Co-occurrence Structure of Cognitive Friction

Beyond individual feature frequency, cognitive friction in procedural language emerges through the structural co-occurrence of multiple judgment-relevant cues within single sentences. Co-occurrence analysis examines how conditional, cross-reference, exception, and discretionary features combine to produce localized interpretive demand in procedural text.

As shown in Figure 3, the strongest co-occurrence pattern in the full corpus involves conditional branching and cross-references (n = 47 sentences in the naturally occurring subset), followed by conditional branching and discretionary qualifiers (n = 9) and conditional branching and exception structures (n = 2). This pattern confirms that conditional structures function as the primary anchor of high-friction configurations, concentrating interpretive demand when combined with referential or discretionary cues.

Discussion

The results show that cognitive friction in procedural text emerges from the co-occurrence of judgment-relevant structural features within single procedural statements. Conditional branching, cross-references, discretionary qualifiers, and exception structures each appear independently

across procedural documentation — but structurally compounded interpretive demand concentrates within a smaller subset of sentences where these features interact. Conditional branching functions as the primary structural anchor: interpretive demand intensifies when conditionality co-occurs with cross-referential or discretionary cues, producing sentences that simultaneously specify contingent execution conditions while deferring decision criteria to external references or actor judgment. This pattern holds consistently across both the full corpus and the 400 naturally occurring sentences, confirming that structurally concentrated judgment demands are a general property of procedural text architecture rather than an artifact of synthetic construction.

Prior STS and organizational accounts have established that documents participate in structuring accountability (Bowker & Star, 1999; Latour, 1987; Suchman, 1987) — but have not specified how this structuring can be observed and measured at the level of sentence architecture. Cognitive friction addresses this gap directly: by operationalizing structural features through rule-based annotation and demonstrating their distribution and co-occurrence patterns across a procedural corpus, this study shows that procedural sentences pre-allocate where actors must resolve ambiguity under accountability-relevant conditions. This is a structural claim about textual design, not a causal claim about downstream attribution outcomes. The link between high-friction configurations and responsibility attribution remains theoretically motivated but empirically open — a direction for future work connecting textual structure to situated interpretive practice.

Within the broader research program on responsibility in regulated environments, cognitive friction specifies the procedural stage of responsibility production: the point at which accountability expectations become operationalized within the architecture of procedural language during execution. Whereas anticipated accountability convergence (AAC) examines how responsibility-relevant judgment conditions are structured before action, and responsibility convergence examines how attribution stabilizes through investigative processes following events, cognitive friction locates where these conditions are pre-configured within procedural documentation itself. These findings have direct implications for the design of procedural documents and interactive systems through which actors navigate instructions under accountability constraints: high-friction configurations — particularly those combining conditional branching with cross-references or discretionary qualifiers — represent design decisions that concentrate interpretive burden at specific points of execution while preserving individual accountability for resulting action. Recognizing cognitive friction as a structural property of procedural inscription, rather than an individual cognitive limitation, shifts the locus of intervention from actor training to document design — with direct consequences for how organizations and system designers configure the interpretive conditions under which accountability is produced.

Limitations and Scope Conditions

Several limitations delimit the scope of the present analysis and indicate directions for further empirical development.

First, although the empirical corpus draws on multiple forms of publicly available procedural documentation — including laboratory guidance, biosafety protocols, chemical safety procedures, and synthetic SOP-like constructions — it remains restricted to highly formalized procedural genres. Such documents provide analytically clear instances of rule-governed inscription, but cognitive friction as a structural property of procedural language may vary across organizational domains with different documentation traditions, levels of formalization, or accountability regimes. Future research may therefore examine how cognitive friction manifests in other procedural environments, including clinical, engineering, or administrative documentation systems.

Second, the present study focuses on sentence-level structural features as minimal units of procedural inscription. This analytical choice enables identification of localized interpretive demands but does not capture higher-order document structures such as section organization, hierarchical nesting of procedures, or cross-document architectures that may also contribute to distributed judgment demands. Extending cognitive friction analysis to multi-sentence or document-level configurations would provide a more comprehensive account of how procedural texts structure interpretive work across organizational documentation systems.

Third, the rule-based annotation procedure operationalizes cognitive friction through explicit lexical and syntactic criteria, which ensures reproducibility and eliminates coder variance. This approach necessarily involves boundary cases — particularly for discretionary qualifiers, where contextual interpretation shapes whether a given construction allocates genuine actor-facing latitude or functions as formulaic hedging. The annotation scripts and illustrative sentence subset are publicly available to support transparency and replication, but future work may benefit from complementing rule-based detection with qualitative validation of boundary cases.

Fourth, the coding framework identifies structural features associated with interpretive demand but does not measure cognitive effort or decision difficulty directly at the level of actors. Cognitive friction is conceptualized here as a property of procedural text rather than individual cognition; however, linking structural configurations to observed interpretive processes or performance outcomes would strengthen the empirical grounding of the framework. Future studies combining textual analysis with observational or experimental approaches could examine how structurally embedded judgment demands are experienced and resolved in situated practice.

Finally, the present analysis does not aim to establish causal relationships between procedural design and downstream responsibility attribution outcomes. While the framework specifies how procedural architecture allocates localized interpretive resolution to actors, accountability

stabilization remains shaped by organizational evaluation processes, institutional norms, and interactional dynamics beyond the text itself. Cognitive friction therefore captures one upstream component of responsibility structuring within sociotechnical systems rather than a complete explanatory account.

Future Research Directions

The cognitive friction framework opens several avenues for further empirical and theoretical development.

One immediate direction involves extending analysis beyond sentence-level configurations to examine how multi-sentence structures and document architectures distribute interpretive demand across procedural systems. Procedural texts often embed decision criteria through hierarchical nesting, section-level dependencies, and cross-document linkages that cannot be fully captured at the level of isolated sentences. Investigating these higher-order textual structures would enable a more comprehensive account of how documentation organizes judgment across organizational knowledge infrastructures.

A second direction concerns variation in cognitive friction across procedural domains and institutional contexts. Comparative analysis across laboratory, clinical, engineering, or administrative procedural systems could reveal how domain-specific documentation conventions shape the distribution and intensity of structurally embedded judgment demands — and whether the conditional-anchored co-occurrence pattern identified here generalizes across institutional genres or reflects properties specific to compliance-oriented procedural documentation.

Third, future work may examine how actors experience and resolve cognitive friction in situated practice. Observational or experimental studies could investigate how practitioners navigate conditional structures, deferred criteria, and cross-references during task execution, and how such navigation shapes decision trajectories and responsibility attribution. Linking textual configurations to observed interpretive processes would deepen the sociotechnical grounding of the framework and directly connect document-level structural analysis to the accountability outcomes it is theorized to shape.

Finally, computational approaches offer opportunities to scale cognitive friction analysis across large document corpora. Automated detection of structural features such as conditionality, cross-referencing, exception handling, and discretionary qualifiers could enable comparative mapping of friction distributions across procedural systems — examining how documentation architectures evolve over time, respond to regulatory change, or vary across organizations.

Conclusion

This study demonstrates that responsibility-relevant judgment conditions in regulated organizational work are partly structured within procedural sentence architecture. Analyzing sentence-level configurations of conditional branching, cross-references, exception structures, and discretionary qualifiers across a 500-sentence corpus — with core findings confirmed across the 400 naturally occurring sentences independently — shows that interpretive demands are not solely products of individual cognition but are partially produced through the architecture of procedural language itself.

The contribution is not to argue that documents shape accountability, but to demonstrate that this shaping is observable and measurable at the level of procedural sentence structure. Cognitive friction renders structurally embedded judgment demands analytically tractable: conditional branching functions as the primary structural anchor of high-friction configurations, concentrating interpretive demand when combined with cross-referential or discretionary cues — and this pattern holds consistently across naturally occurring procedural documentation spanning biosafety, laboratory, and chemical safety genres.

Cognitive friction specifies the procedural stage at which accountability expectations become operationalized within procedural language during execution — linking anticipatory alignment to investigative stabilization through the architecture of procedural text itself. Future research may extend this framework across domains and document architectures, and examine how procedural structuring interacts with anticipatory alignment, investigative stabilization, and AI-mediated representation across responsibility lifecycles.

Code Availability

The code supporting this study is publicly available at: `https://github.com/sorakim-lab/sop-cognitive-friction-pilot`

The repository contains the rule-based annotation scripts and analysis workflow used to identify sentence-level cognitive friction features in the 500-sentence procedural corpus analyzed in this study, including robustness analyses conducted on the 400 naturally occurring sentences independently of the synthetic subset.

Data Availability

The procedural sentence corpus analyzed in this study consists of 500 procedural sentences derived from publicly accessible regulatory and safety documentation, including WHO laboratory guidance, academic laboratory safety resources, and related procedural materials. Because the

corpus includes sentences adapted from publicly available documents and synthetic constructions, only an illustrative subset and annotation outputs are provided in the public repository. The full annotated corpus is available from the author upon reasonable request.

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